Ratio Reasoning: Convert Measurements within the Same System

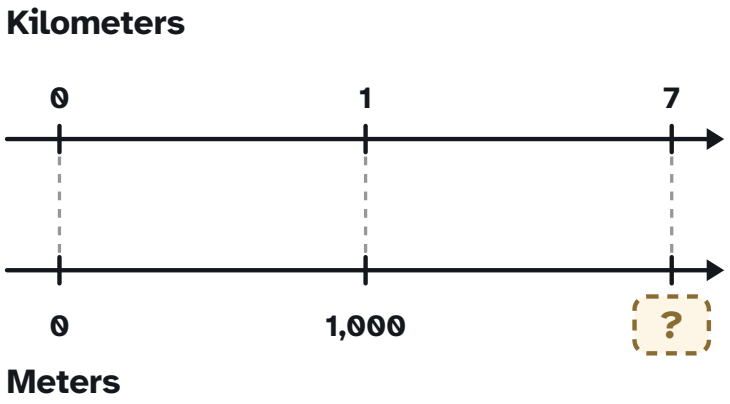
Convert metric units with powers of ten and check that the answer makes sense.

Name _____ **Date** _____

Tonight's spark

A team cooler holds 6 liters of water. The coach's cups are marked in milliliters. One liter is 1,000 milliliters. How many milliliters does the cooler hold?

Make a guess first: is it closer to 60, 600 or 6,000?



Our class model: the power of ten jump

Try it

Work it out on paper. Write one answer on each line.

1 **How many meters are in 1 kilometer?**

A. 10 **B.** 100 **C.** 1,000 **D.** 10,000

My answer _____

2 **5 kilometers = ? meters**

A. 5,000 **B.** 50 **C.** 500 **D.** 0.005

My answer _____

3 **Change 3,000 grams to kilograms. (1 kilogram = 1,000 grams)**

A. 30 **B.** 3 **C.** 300 **D.** 3,000,000

My answer _____

4 **How many centimeters are in 4 meters? (1 meter = 100 centimeters)**

A. 0.04 **B.** 40 **C.** 404 **D.** 400

My answer _____

5 **Convert 2 liters to milliliters.**

A. 2,000 **B.** 200 **C.** 20 **D.** 0.002

My answer _____

6 **Convert 650 millimeters to meters. (1 meter = 1,000 millimeters)**

A. 650,000 **B.** 6.5 **C.** 0.65 **D.** 65

My answer _____

Talk about it

Are you changing to a smaller unit or a bigger unit, and what does that tell you to do?

Home mission

Find a bottle labeled 1 L or 1,000 mL. Say: "The cooler holds 6 liters." Ask your student how many bottles like this fill it. Then ask how many milliliters that is in all.

If we get stuck, ask

- Which unit is smaller — the one we have, or the one we want?
- Will there be more of them or fewer of them?
- Does the answer look reasonable for that unit?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/3-6-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it